2Market

Data Analytics

By

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2 Market Data Analysis

The Business Problem

2Market are a global supermarket chain with both physical stores in eight countries and an online presence. They wish to understand their customers better, especially regarding demographics.

The analytics project will identify what 2Market's customer demographic is, which advertising channels are most successful, as well as those product categories that sell the best and whether customer demographics play a part in product sales.

The analysis was completed using the IDEAL framework developed by Bransford & Stein (1984)

1. Identify the problem & approaches
   - 2Market business problem defined above
2. Define goals
   - To understand 2Market customer demographics
3. Explore possible solutions & strategies
   - exploratory in Excel, Tableau & SQL
4. Anticipate outcomes and act on them
   - Using past sales data and advertising to predict outcomes
5. Look back and learn
   - 2Market should continue to review its customer base and the effectiveness of advertising

The stakeholders interested in this analysis are likely to be executive board members who make the decisions on how to move 2Market forward.

Cleaning & Analysing the Data

The dataset as a CSV file was imported into Excel to allow cleaning, as follows:

- The date column was formatted to DD/MM/YYYY
- Duplicates. The primary key (ID column) must be unique. There were no duplicate PKs. However, glancing over the data revealed there appears to be rows that contain the exact same entities except for 'PK' and 'country', or 'PK' and 'Response'. These rows were deleted, as it very unlikely that all the other entities would be identical.
- Spelling and standardisation regarding capital and lower-case letters were checked.
- Blank cells were identified. None found.
- The '$' and ', ' were removed from the income figures.
- Spaces which may or may not have been present, were removed using the TRIM function.
- To aid the analysis two new columns were added. 'Age', next to 'Year_Birth', calculated as current_year - 'Year_Birth;, and 'Total_Spend', the sum of all product categories per customer.
- The product categories were changed to more meaningful titles ie 'AmtLiq' to 'Alcohol'.

Sorting the 'Year_Birth' column by lowest to highest reveals three abnormalities. These entries suggested customer ages over 124, which is implausible and therefore removed

An observation from the birth year is that the youngest customer would have been only fifteen when they first registered with 2Market.

The 'Marital status' column threw up a couple of oddities, namely 'YOLO' and 'Absurd'. These could have been combined into a new selection "Unknown", but, as there were only three entries, they wouldn't show in the analysis, so those were removed.

Similarly, there are categories for 'Single' and 'Alone', these were merged as 'Single'.

Ensured consistent data types using Excel's TYPE function

Noted the outlier in income, column of $666,666, way above the second highest income. Left in for the initial analysis but later removed for analysis involving income as it would affect the average, although the median could be used instead of average to get around this.

The income range was from $1730 to $162,397 with an average of $52,026. Incomes seems to be both annual and monthly. No changes were made.

The Excel analysis revealed:

- The age of customers ranges from 28 to 84, with the average being 55. Married couples make up the largest group of customers. This was plotted by a line chart (fig 1) and bar chart (fig 2), respectively.
- Income was analysed by a scatter plot (fig 3), which shows the bulk of customers have an income of between $20k and $100k.
- Spain provides the highest amount of revenue but also has the largest customer base. Montenegro has the lowest.
- Germany has the highest average spend per customer, while India the lowest. Montenegro an outlier, was removed for this analysis.

Turning to SQL, a new database was created in Postgres SQL. Two tables were made, and the cleaned data was imported.

The SQL analysis revealed:

- Spain produced the highest revenue by far over three times more than the next highest which is South Africa. Montenegro lags way behind
- Alcohol is by far the best selling product category across the board accounting for 50% of all sales, with vegetables and chocolate being the least popular (fig 4). Interestingly this doesn't change by marital status or if kids or teens are at home.
- The most successful social media platforms to advertise on in order are Twitter, Instagram then Facebook.
- Interestingly bulk email was popular amongst single people, nearly as much as Twitter, while brochure ads were unpopular across the board.

Dashboard Design & Development

The dashboard was planned using pen and paper to put some ideas together and get an idea of the layout. After several prototype builds, I decided on a calm blue colour scheme, which is appealing and easy to read.

The dashboard was built from visualisations created in Tableau to analysing demographics and spending behaviour. They specifically show:

- Age range of the customers
- Marital status
- Education level
- A world map showing the geographical location and density of customers
- Income and number of customers in each income bracket
- Total sales by product category
- Success of advertising campaigns

Insights and Trends

2Market wanted to understand the key demographics of its customers, best selling products, if demographics play a part and which advertising channels seem the most effective.

Customer Demographics

2Market operate in eight countries around the world. Customers range in age from 28 to 84, the average age is 55, with the majority either married or together. Most are educated to graduate level or above.

Spain has by far the largest number of customers and as such also produces the highest revenue. Montenegro is the worst performing with only two customers. However, Germany produced the largest average spend per customer (fig 5).

Product Sales

Alcohol is by far the best selling product category, accounting for 50% of all sales, followed by meat. On the other hand, vegetable sales are the worst. This trend is repeated for all countries.

Advertising Channels

Twitter is the most effective advertising platform, followed by Instagram and Facebook. Bulk email is surprisingly effective particularly amongst singletons. Brochure advertising is the worst performing (fig 6).

Recommendations

- Concentrate on the core revenue providers of alcohol and meat.
- Continue to use social media platforms form advertising, particularly Twitter, drop brochure advertising. Could also target singletons with bulk email.
- Drop markets where revenue is low, particularly Montenegro or maybe it's a market to develop
- Maybe target post graduation level of education customers

Appendix 1 Charts & Graphs

Fig 1: Minimum, maximum, and average age by marital status

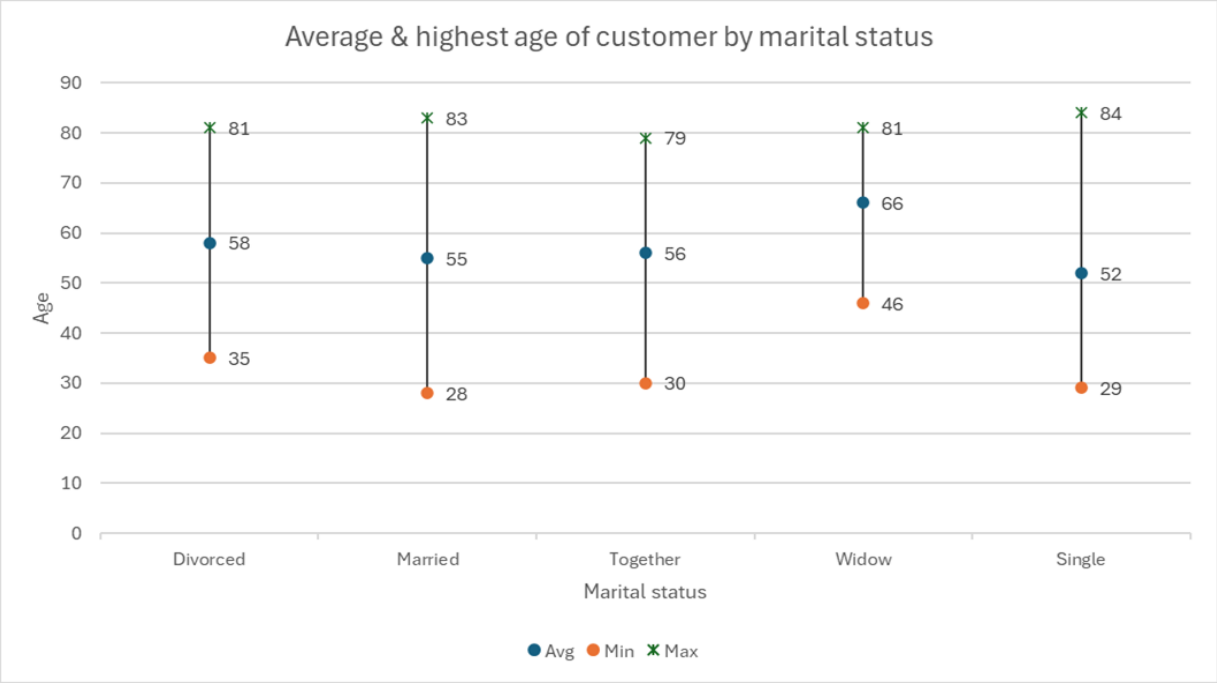


Fig 2: Customer count by country

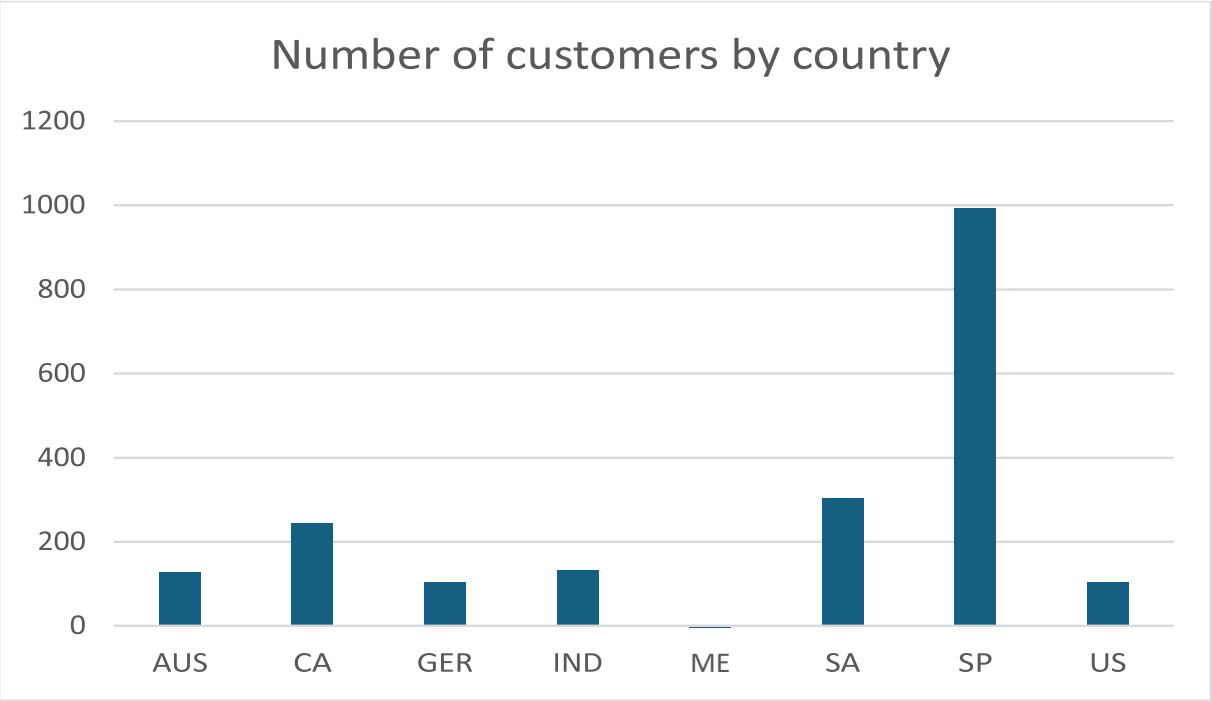


Fig 3: Income by age grouping

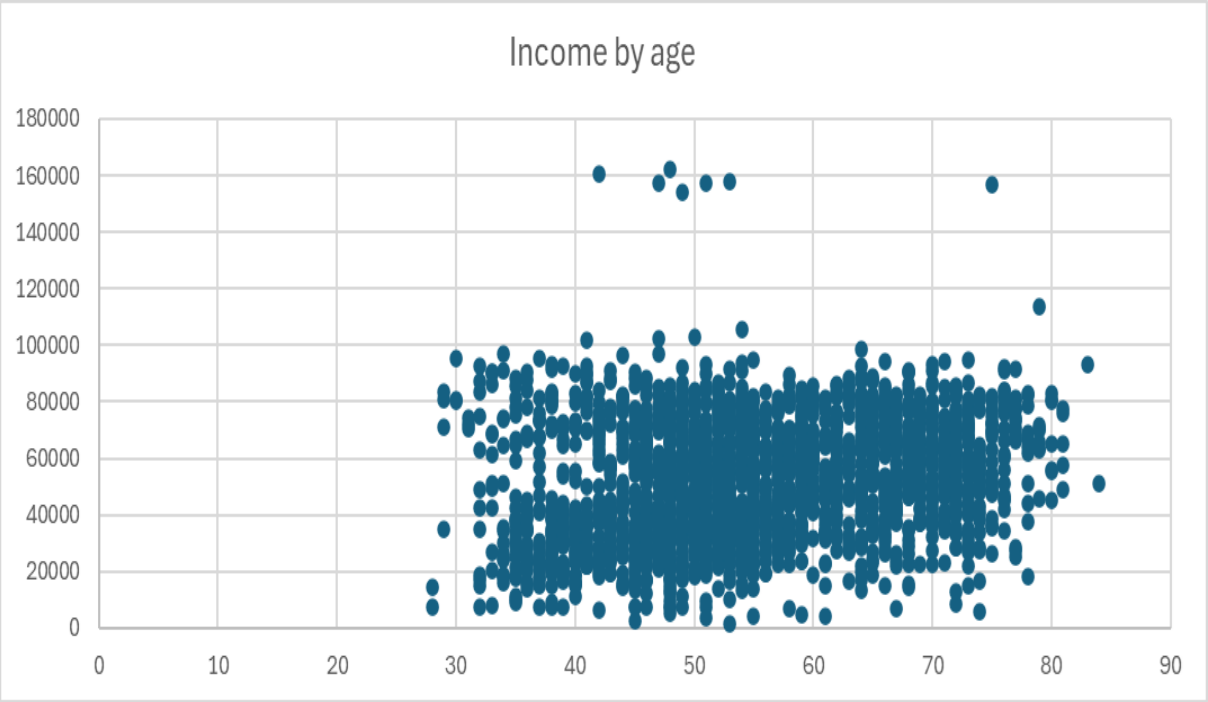


Fig 4: Best selling product categories

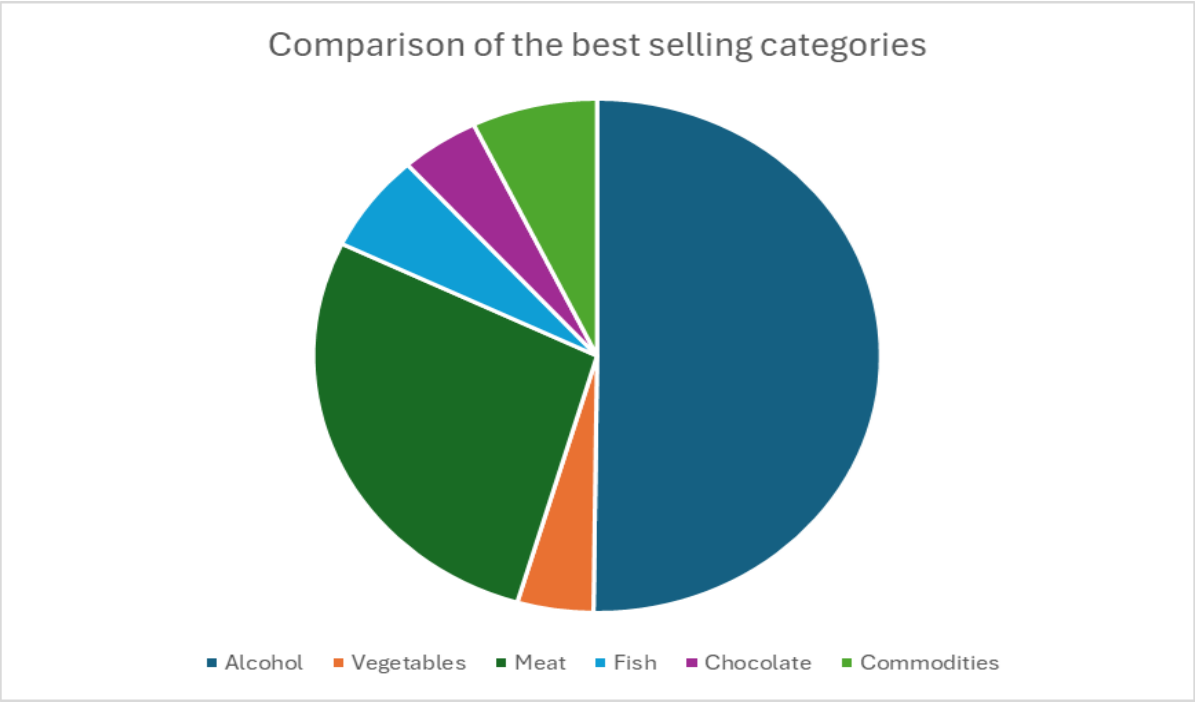
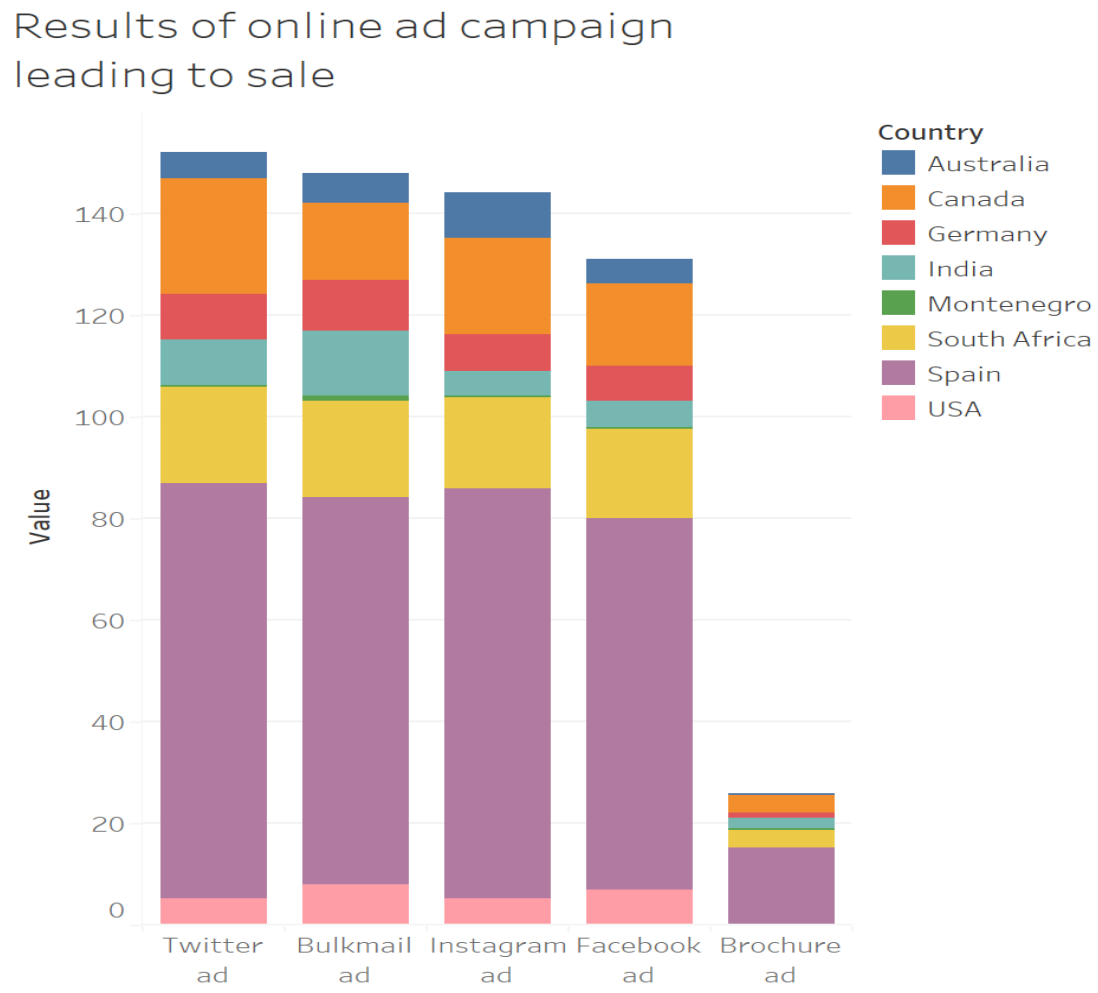


Fig 5: Average customer spend by country

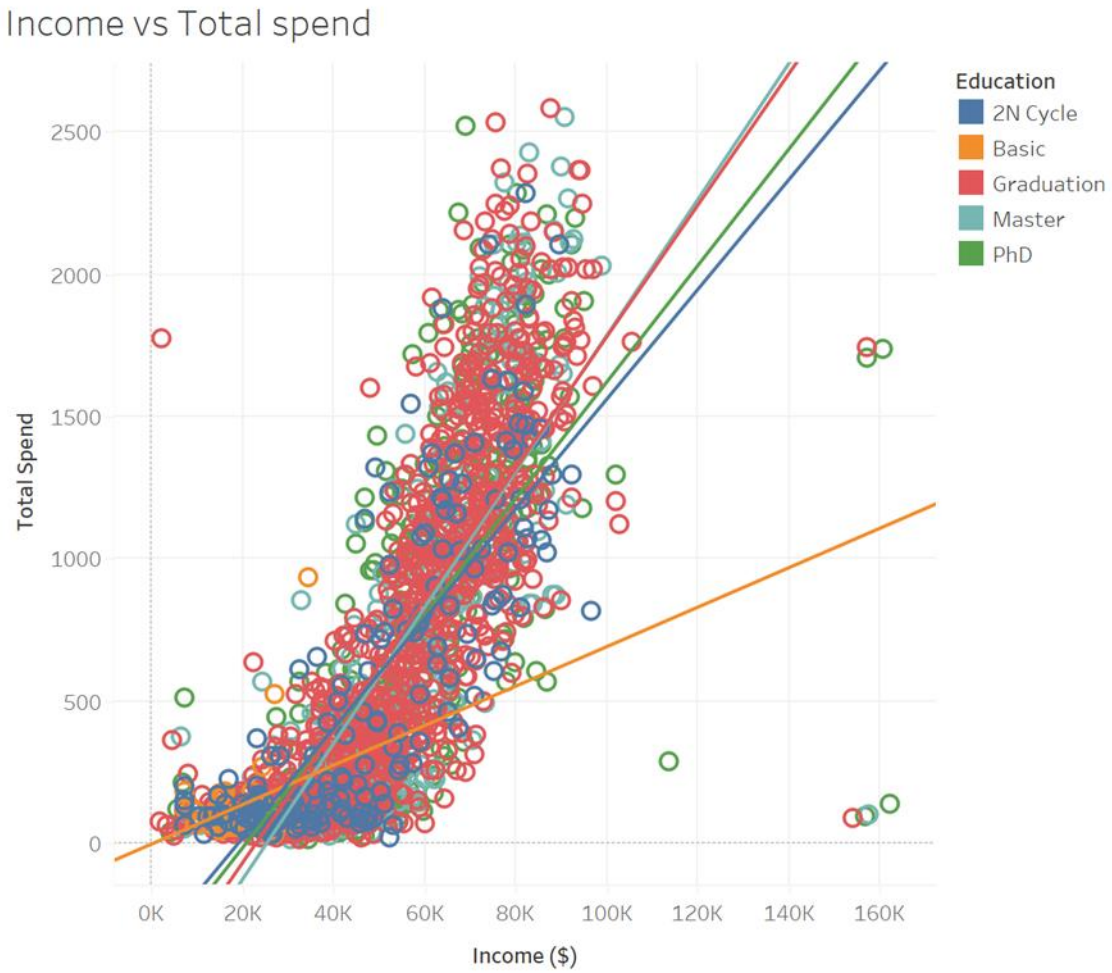
Country	Avg spend	Total spend	No of cust
Spain	$656.01	$652,074.00	994
South Africa	$674.52	$204,380.00	303
Canada	$670.08	$163,499.00	244
India	$580.45	$76,619.00	132
Australia	$598.80	$76,048.00	127
Germany	$694.50	$71,533.00	103
USA	$689.07	$70,974.00	103
Montenegro	$1,074.50	$2,149.00	2

Fig 6: Advertising campaign success by country



Twitter ad, Bulkmail ad, Instagram ad, Facebook ad and Brochure ad. Colour shows details about Country. The view is filtered on Country, which keeps 8 of 8 members.

Fig 7: Income vs total spending



Income ($) vs. Total Spend. Colour shows details about Education. The data is filtered on Income ($), which ranges from 1730 to 163000. The view is filtered on Education, which keeps 2N Cycle, Basic, Graduation, Master and PhD.

Appendix 2: Excel Data set

Marketing dataset (Cleaned)

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA
1	ID	Year_Birth	Age	Education	Marital_Status	Income_($)	Kidhome	Teenhome	Dt_Customer	Recency	Alcohol	Vegetables	Meat	Fish	Chocolate	Commodities	Total_Spend	NumDeals	NumWebBuy	NumWalkinPur	NumVisits	Response	Complain	Country	Count_success		
2	6862	1972	53	Graduation	Divorced	1730	0	0	18/05/2014	65	1	1	3	1	1	1	73	15	0	0	20	0	0	SP	0		
3	5376	1980	45	Graduation	Married	2447	1	0	06/01/2013	42	1	1	1725	1	1	1	1772	15	0	0	1	0	0	US	0		
4	11110	1974	51	Graduation	Single	3502	1	0	13/04/2013	56	2	1	1	0	0	1	61	0	0	0	14	0	0	IND	0		
5	9931	1964	61	PhD	Married	4023	1	1	23/06/2014	29	5	0	1	1	1	1	38	15	0	0	19	0	0	SP	0		
6	10311	1970	55	Graduation	Married	4428	0	1	05/10/2013	0	16	4	12	2	4	321	359	0	25	0	1	0	0	SP	0		
7	3955	1966	59	Graduation	Divorced	4861	0	0	22/06/2014	20	2	1	1	1	0	1	26	0	0	0	14	0	0	SA	0		
8	9303	1977	48	Graduation	Married	5305	0	1	30/07/2013	12	12	4	7	1	3	5	44	0	1	0	13	0	0	IND	0		
9	5899	1951	74	PhD	Together	5648	1	1	13/03/2014	68	28	0	6	1	1	13	117	1	1	0	20	0	0	US	1		
10	4246	1983	42	Master	Single	6560	0	0	12/12/2013	2	67	11	26	4	3	262	375	0	1	1	17	0	0	SP	0		
11	4303	1958	67	PhD	Together	6835	0	1	08/12/2012	76	107	2	12	2	2	12	213	0	0	1	20	0	0	CA	0		
12	6237	1967	58	PhD	Single	7144	0	2	07/12/2013	92	81	4	33	5	2	291	508	0	23	1	0	0	0	SP	0		
13	9909	1997	28	2N Cycle	Married	7500	0	0	09/11/2012	24	3	18	14	15	22	50	146	3	3	3	9	1	0	AUS	0		
14	4136	1993	32	Basic	Single	7500	1	0	03/12/2012	63	10	17	18	8	26	40	182	4	3	2	9	0	0	SP	0		
15	9553	1988	37	Graduation	Married	7500	0	0	09/01/2013	94	0	2	3	4	1	5	109	1	0	3	6	0	0	IND	0		
16	456	1987	38	2N Cycle	Married	7500	1	0	07/02/2013	96	1	11	5	4	6	9	132	2	2	3	8	0	0	SP	0		
17	10001	1986	39	2N Cycle	Together	7500	1	0	01/08/2012	98	5	17	17	13	14	34	198	4	2	3	9	0	0	SP	0		
18	10659	1980	45	2N Cycle	Together	7500	1	0	07/05/2013	7	2	8	11	3	8	21	60	4	3	2	7	0	0	IND	0		
19	10710	1980	45	Graduation	Married	7500	0	1	29/08/2012	61	5	2	3	3	0	5	79	1	1	2	8	0	0	SP	0		
20	10641	1979	46	Graduation	Single	7500	1	1	27/12/2012	5	6	5	4	13	4	25	62	4	2	3	6	0	0	SP	0		
21	4692	1977	48	Graduation	Married	7500	1	0	01/08/2012	19	7	0	12	13	7	32	90	5	4	2	9	1	0	AUS	1		
22	387	1977	48	Basic	Married	7500	0	0	13/11/2012	59	6	16	11	11	1	16	120	1	2	3	8	0	0	US	0		
23	2139	1976	49	Master	Married	7500	1	0	02/10/2013	19	3	1	10	3	2	12	50	3	2	3	5	0	0	AUS	0		
24	7297	1974	51	Graduation	Single	7500	1	0	26/12/2013	54	5	3	10	12	7	20	111	4	3	3	7	0	0	SA	0		
25	10749	1992	33	Graduation	Single	8028	0	0	18/09/2012	62	73	18	66	7	12	2	240	15	0	0	19	0	0	SP	0		
26	2980	1953	72	Master	Single	8820	1	1	22/03/2013	52	12	0	13	4	2	4	87	4	3	3	8	0	0	CA	0		
27	10395	1987	38	Basic	Single	8940	1	0	22/08/2012	25	1	45	3	4	6	42	126	3	3	3	8	0	0	SP	0		
28	451	1990	35	Graduation	Single	9255	1	0	20/12/2013	91	9	0	7	2	0	4	113	1	1	2	8	0	0	IND	0		
29	5342	1977	48	Basic	Divorced	9548	1	0	08/08/2012	31	0	1	3	10	6	9	60	2	1	3	8	0	0	CA	0		

Ad Dataset (Cleaned)

ID	Bulkmail_a	Twitter_ad	Instagram_	Facebook_a	Brochure_ad
1826	0	0	0	0	0
1	0	0	0	0	1
10476	0	0	0	0	0
1386	0	0	0	0	0
5371	1	0	0	0	0
7348	0	0	0	0	0
4073	1	0	0	0	0
1991	0	0	0	0	0
4047	0	0	0	0	0
9477	0	0	0	0	0
2079	0	0	0	0	0
5642	0	0	0	0	0
10530	0	0	0	0	0
2964	0	0	0	0	0
10311	0	0	0	0	0
837	0	0	0	0	0
10521	0	0	0	0	0
10175	0	0	0	0	0
1473	0	0	0	0	0
2795	0	0	0	0	0
2285	0	0	0	0	0
115	0	0	0	0	0
10470	0	0	0	0	0

Appendix 3: SQL scripts

Create the 2Market Database

```
-- Create Database--
CREATE DATABASE "2Market";
```

Create the marketing data table

```
--Create table Marketing Data--
CREATE TABLE "Marketing_Data" (
  "ID" INTEGER PRIMARY KEY,
  "Year_Birth" NUMERIC(4,0),
        "Age" NUMERIC(3,0),
        "Education" VARCHAR(20),
        "Marital_Status" VARCHAR(20),
        "Income_($)" NUMERIC(6,0),
        "Kidhome" NUMERIC(1,0),
        "Teenhome" NUMERIC(1,0),
        "Dt_Customer" DATE,
        "Recency" NUMERIC(2,0),
        "Alcohol" NUMERIC(4,0),
        "Vegetables" NUMERIC(4,0),
        "Meat" NUMERIC(4,0),
        "Fish" NUMERIC(4,0),
        "Chocolate" NUMERIC(4,0),
        "Commodities" NUMERIC(4,0),
        "Total_Spend" NUMERIC(8,0),
        "NumDeals" NUMERIC(2,0),
        "NumWebBuy" NUMERIC(4,0),
```

```
        "NumWalkinPur" NUMERIC(2,0),
        "NumVisits" NUMERIC(2,0),
        "Response" BOOLEAN,
        "Complain" BOOLEAN,
        "Country" VARCHAR(20),
        "Count_success" NUMERIC(1,0)
);
```

Create the ad data table

```
-- Create Table "ad_data"--
CREATE TABLE "ad_data" (
  "ID" INTEGER PRIMARY KEY,
        "Bulkmail_ad" BOOLEAN,
        "Twitter_ad" BOOLEAN,
        "Instagram_ad" BOOLEAN,
        "Facebook_ad" BOOLEAN,
        "Brochure_ad" BOOLEAN);
```

Import the data into the tables (Marketing data)

Data Output Messages Notifications

Showing rows: 1 to 1000 Page No: 1 of 3

	ID [PK] integer	Year_Birth numeric (4)	Age numeric (3)	Education character varying (20)	Marital_Status character varying (20)	Income_($) numeric (6)	Kidhome numeric (1)	Teenhome numeric (1)	Dt_Customer date	Recency numeric (2)	Alcohol numeric (4)
1	6862	1972	53	Graduation	Divorced	1730	0	0	2014-05-18	65	
2	5376	1980	45	Graduation	Married	2447	1	0	2013-01-06	42	
3	11110	1974	51	Graduation	Single	3502	1	0	2013-04-13	56	
4	9931	1964	61	PhD	Married	4023	1	1	2014-06-23	29	
5	10311	1970	55	Graduation	Married	4428	0	1	2013-10-05	0	1
6	3955	1966	59	Graduation	Divorced	4861	0	0	2014-06-22	20	
7	9303	1977	48	Graduation	Married	5305	0	1	2013-07-30	12	1
8	5899	1951	74	PhD	Together	5648	1	1	2014-03-13	68	2

Data Output Messages Notifications

Showing rows: 1 to 1000 Page No: 1 of 3

	Alcohol numeric (4)	Vegetables numeric (4)	Meat numeric (4)	Fish numeric (4)	Chocolate numeric (4)	Commodities numeric (4)	Total_Spend numeric (8)	NumDeals numeric (2)	NumWebBuy numeric (4)	NumWalkinPur numeric (2)	NumVisits numeric (2)	Response boolean
1	65	1	1	3	1	1	1	73	15	0	0	20	false
2	42	1	1	1725	1	1	1	1772	15	0	0	1	false
3	56	2	1	1	0	0	1	61	0	0	0	14	false
4	29	5	0	1	1	1	1	38	15	0	0	19	false
5	0	16	4	12	2	4	321	359	0	25	0	1	false
6	20	2	1	1	1	0	1	26	0	0	0	14	false
7	12	12	4	7	1	3	5	44	0	1	0	13	false
8	68	28	0	6	1	1	13	117	1	1	0	20	false

Data Output Messages Notifications

Showing rows: 1 to 1000 Page No: 1 of 3

	)	Chocolate numeric (4)	Commodities numeric (4)	Total_Spend numeric (8)	NumDeals numeric (2)	NumWebBuy numeric (4)	NumWalkinPur numeric (2)	NumVisits numeric (2)	Response boolean	Complain boolean	Country character varying (20)	Count_success numeric (1)
1	1	1	1	73	15	0	0	20	false	false	SP	0
2	1	1	1	1772	15	0	0	1	false	false	US	0
3	0	0	1	61	0	0	0	14	false	false	IND	0
4	1	1	1	38	15	0	0	19	false	false	SP	0
5	2	4	321	359	0	25	0	1	false	false	SP	0
6	1	0	1	26	0	0	0	14	false	false	SA	0
7	1	3	5	44	0	1	0	13	false	false	IND	0
8	1	1	13	117	1	1	0	20	false	false	US	1

Import the data into the table (ad data)

Data Output Messages Notifications

	ID [PK] integer	Bulkmail_ad boolean	Twitter_ad boolean	Instagram_ad boolean	Facebook_ad boolean	Brochure_ad boolean
1	1826	false	false	false	false	false
2	1	false	false	false	false	true
3	10476	false	false	false	false	false
4	1386	false	false	false	false	false
5	5371	true	false	false	false	false
6	7348	false	false	false	false	false
7	4073	true	false	false	false	false
8	1991	false	false	false	false	false
9	4047	false	false	false	false	false

The Total spend per country

```
--Total spend per country---
SELECT "Country", SUM("Total_Spend") AS "Revenue"
  FROM public."Marketing_Data"
GROUP BY "Country"
ORDER BY "Revenue" DESC;
```

Data Output Messages Notifications

Showing rows: 1 to 8 Page No: 1 of 1

	Country character varying (6)	Revenue bigint
1	SP	652074
2	SA	204380
3	CA	163499
4	IND	76619
5	AUS	76048
6	GER	71533
7	US	70974
8	ME	2149

The total spend per product by country

```
--Total spend per product per country--
SELECT SUM("Alcohol") AS "Alcohol",
    SUM("Chocolate") AS "Chocolate",
        SUM("Commodities") AS "Commodities",
        SUM("Fish") AS "Fish",
        SUM("Meat") AS "Meat",
        SUM("Vegetables") AS "Vegetables",
        "Country"
FROM public."Marketing_Data"
GROUP BY "Country"
ORDER BY "Country";
```

Showing rows: 1 to 8 Page No: 1

	Alcohol bigint	Choclate bigint	Commodities bigint	Fish bigint	Meat bigint	Vegetables bigint	Country character varying (6)
1	34377	3444	5885	4692	19011	3048	AUS
2	77316	6945	10868	8627	41421	6900	CA
3	33032	2571	5088	4210	18712	2855	GER
4	32651	2919	5208	4334	21410	3388	IND
5	940	113	182	214	675	8	ME
6	95353	7981	13809	12405	51921	8063	SA
7	307188	27857	41598	36379	163763	25693	SP
8	31084	2805	4694	4320	19967	2927	US

Which products are most popular in each country

```
-.Which products are most popular in each country--
SELECT
  "Country",
  CASE
    WHEN GREATEST(SUM("Alcohol"), SUM("Chocolate"), SUM("Commodities"), SUM("Fish"), SUM("Meat"), SUM("Vegetables")) = SUM("Alcohol") THEN 'Alcohol'
    WHEN GREATEST(SUM("Alcohol"), SUM("Chocolate"), SUM("Commodities"), SUM("Fish"), SUM("Meat"), SUM("Vegetables")) = SUM("Chocolate") THEN 'Chocolate'
    WHEN GREATEST(SUM("Alcohol"), SUM("Chocolate"), SUM("Commodities"), SUM("Fish"), SUM("Meat"), SUM("Vegetables")) = SUM("Commodities") THEN 'Commodities'
    WHEN GREATEST(SUM("Alcohol"), SUM("Chocolate"), SUM("Commodities"), SUM("Fish"), SUM("Meat"), SUM("Vegetables")) = SUM("Fish") THEN 'Fish'
    WHEN GREATEST(SUM("Alcohol"), SUM("Chocolate"), SUM("Commodities"), SUM("Fish"), SUM("Meat"), SUM("Vegetables")) = SUM("Meat") THEN 'Meat'
    WHEN GREATEST(SUM("Alcohol"), SUM("Chocolate"), SUM("Commodities"), SUM("Fish"), SUM("Meat"), SUM("Vegetables")) = SUM("Vegetables") THEN 'Vegetables'
  END AS "Most_Popular_Product"
FROM
  public."Marketing_Data"
GROUP BY
  "Country"
ORDER BY
  "Country";
```

Data Output Messages Notifications

Showing rows: 1 to 8

	Country character varying (6)	Most_Popular_Product text
1	AUS	Alcohol
2	CA	Alcohol
3	GER	Alcohol
4	IND	Alcohol
5	ME	Alcohol
6	SA	Alcohol
7	SP	Alcohol
8	US	Alcohol

Which products are the most popular based on marital status

```
--Which products are most popular based on marital status--

SELECT

  "Marital_Status",

  CASE

        --Adding the totals of each product category and finding the highest number--

    WHEN GREATEST(SUM("Alcohol"), SUM("Chocolate"), SUM("Commodities"), SUM("Fish"), SUM("Meat"), SUM("Vegetables")) = SUM("Alcohol") THEN 'Alcohol'

    WHEN GREATEST(SUM("Alcohol"), SUM("Chocolate"), SUM("Commodities"), SUM("Fish"), SUM("Meat"), SUM("Vegetables")) = SUM("Chocolate") THEN 'Chocolate'

    WHEN GREATEST(SUM("Alcohol"), SUM("Chocolate"), SUM("Commodities"), SUM("Fish"), SUM("Meat"), SUM("Vegetables")) = SUM("Commodities") THEN 'Commodities'

    WHEN GREATEST(SUM("Alcohol"), SUM("Chocolate"), SUM("Commodities"), SUM("Fish"), SUM("Meat"), SUM("Vegetables")) = SUM("Fish") THEN 'Fish'

    WHEN GREATEST(SUM("Alcohol"), SUM("Chocolate"), SUM("Commodities"), SUM("Fish"), SUM("Meat"), SUM("Vegetables")) = SUM("Meat") THEN 'Meat'

    WHEN GREATEST(SUM("Alcohol"), SUM("Chocolate"), SUM("Commodities"), SUM("Fish"), SUM("Meat"), SUM("Vegetables")) = SUM("Vegetables") THEN 'Vegetables'

  END AS "Most_Popular_Product",

 CASE

  --Adding the totals of each product category add finding the lowest number--

    WHEN LEAST(SUM("Alcohol"), SUM("Chocolate"), SUM("Commodities"), SUM("Fish"), SUM("Meat"), SUM("Vegetables")) = SUM("Alcohol") THEN 'Alcohol'
```

```
    WHEN LEAST(SUM("Alcohol"), SUM("Chocolate"), SUM("Commodities"),
SUM("Fish"), SUM("Meat"), SUM("Vegetables")) = SUM("Chocolate") THEN 'Chocolate'

    WHEN LEAST(SUM("Alcohol"), SUM("Chocolate"), SUM("Commodities"),
SUM("Fish"), SUM("Meat"), SUM("Vegetables")) = SUM("Commodities") THEN
'Commodities'

    WHEN LEAST(SUM("Alcohol"), SUM("Chocolate"), SUM("Commodities"),
SUM("Fish"), SUM("Meat"), SUM("Vegetables")) = SUM("Fish") THEN 'Fish'

    WHEN LEAST(SUM("Alcohol"), SUM("Chocolate"), SUM("Commodities"),
SUM("Fish"), SUM("Meat"), SUM("Vegetables")) = SUM("Meat") THEN 'Meat'

    WHEN LEAST(SUM("Alcohol"), SUM("Chocolate"), SUM("Commodities"),
SUM("Fish"), SUM("Meat"), SUM("Vegetables")) = SUM("Vegetables") THEN 'Vegetables'

  END AS "Least_Popular_Product"

FROM

  public."Marketing_Data"

GROUP BY

  "Marital_Status"

ORDER BY

  "Marital_Status";
```

Data Output Messages Notifications

	Marital_Status character varying (12)	Most_Popular_Product text	Least_Popular_Product text
1	Divorced	Alcohol	Chocolate
2	Married	Alcohol	Vegetables
3	Single	Alcohol	Vegetables
4	Together	Alcohol	Vegetables
5	Widow	Alcohol	Vegetables

Which products are most popular based on kids or teens at home

```
--Which products are most popular based on kids or teens at home--
SELECT
  CASE
--Setting up two arguments, kids or teens at home & kids or teens not at home--
--if kidhome or teenhome>0 then put the totals into 'kids or teens'--
    WHEN "Kidhome" > 0 OR "Teenhome" > 0 THEN 'Kids or Teens'
--if kidhome or teenhome=0 then put totals into 'No kids or teens'--
    ELSE 'No Kids or Teens'
  END AS "Kids or teens",
--Adding up the totals of each product group--
  SUM("Alcohol") AS "Alcohol",
  SUM("Chocolate") AS "Chocolate",
  SUM("Commodities") AS "Commodities",
  SUM("Fish") AS "Fish",
  SUM("Meat") AS "Meat",
  SUM("Vegetables") AS "Vegetables"
FROM
--Using the 'Marketing_Data' table--
  public."Marketing_Data"
--Grouping the results by either 'kids at home' or 'no kids at home' by using the CASE function--
GROUP BY
  CASE
    WHEN "Kidhome" > 0 OR "Teenhome" > 0 THEN 'Kids or Teens'
    ELSE 'No Kids or Teens'
  END
ORDER BY
  "Kids or teens";
```

Data Output Messages Notifications

Showing rows

	Kids or teens text	Alcohol bigint	Chocolate bigint	Commodities bigint	Fish bigint	Meat bigint	Vegetables bigint
1	Kids or Teens	332406	24200	51544	32122	124191	23132
2	No Kids or Teens	279535	30435	35788	43059	212689	29750

Which advertising method is most effective by marital status

```
--5.2 Which ad method is most effective by marital status--
SELECT
   md."Marital_Status",
         --Adding up the sum of each ad and finding the highest number--
         --Boolean (true/false) converted to integer 1/0)--
  CASE
    WHEN GREATEST(SUM("Bulkmail_ad"::INT), SUM("Twitter_ad"::INT),
                  SUM("Instagram_ad"::INT), SUM("Facebook_ad"::INT),
SUM("Brochure_ad"::INT)) = SUM("Bulkmail_ad"::INT) THEN 'Bulkmail'
    WHEN GREATEST(SUM("Bulkmail_ad"::INT), SUM("Twitter_ad"::INT),
                  SUM("Instagram_ad"::INT), SUM("Facebook_ad"::INT::INT),
SUM("Brochure_ad"::INT)) = SUM("Twitter_ad"::INT) THEN 'Twitter'
                  WHEN GREATEST(SUM("Bulkmail_ad"::INT), SUM("Twitter_ad"::INT),
                  SUM("Instagram_ad"::INT), SUM("Facebook_ad"::INT),
SUM("Brochure_ad"::INT)) = SUM("Instagram_ad"::INT) THEN 'Instagram'
                  WHEN GREATEST(SUM("Bulkmail_ad"::INT), SUM("Twitter_ad"::INT),
                  SUM("Instagram_ad"::INT), SUM("Facebook_ad"::INT),
SUM("Brochure_ad"::INT)) = SUM("Facebook_ad"::INT) THEN 'Facebook'
                  WHEN GREATEST(SUM("Bulkmail_ad"::INT), SUM("Twitter_ad"::INT),
                  SUM("Instagram_ad"::INT), SUM("Facebook_ad"::INT),
SUM("Brochure_ad"::INT)) = SUM("Brochure_ad"::INT) THEN 'Brochure'
  END AS "Most_Popular_ad_method",
         --Adding up the sum of each ad and finding the lowst number--
         --Boolean (true/false) converted to integer 1/0)--
```

```
  CASE
        WHEN LEAST(SUM("Bulkmail_ad"::INT), SUM("Twitter_ad"::INT),
              SUM("Instagram_ad"::INT), SUM("Facebook_ad"::INT),
SUM("Brochure_ad"::INT)) = SUM("Bulkmail_ad"::INT) THEN 'Bulkmail'
              WHEN LEAST(SUM("Bulkmail_ad"::INT), SUM("Twitter_ad"::INT),
              SUM("Instagram_ad"::INT), SUM("Facebook_ad"::INT),
SUM("Brochure_ad"::INT)) = SUM("Twitter_ad"::INT) THEN 'Twitter'
              WHEN LEAST(SUM("Bulkmail_ad"::INT), SUM("Twitter_ad"::INT),
              SUM("Instagram_ad"::INT), SUM("Facebook_ad"::INT),
SUM("Brochure_ad"::INT)) = SUM("Instagram_ad"::INT) THEN 'Instagram'
              WHEN LEAST(SUM("Bulkmail_ad"::INT), SUM("Twitter_ad"::INT),
SUM("Instagram_ad"::INT), SUM("Facebook_ad"::INT),
              SUM("Brochure_ad"::INT)) = SUM("Facebook_ad"::INT) THEN 'Facebook'
              WHEN LEAST(SUM("Bulkmail_ad"::INT), SUM("Twitter_ad"::INT),
              SUM("Instagram_ad"::INT), SUM("Facebook_ad"::INT),
SUM("Brochure_ad"::INT)) = SUM("Brochure_ad"::INT) THEN 'Brochure'

  END AS "Least_Popular_ad_method"
FROM "Marketing_Data" md
INNER JOIN "ad_data" ad
ON md."ID" = ad."ID"
GROUP BY
  md."Marital_Status"
ORDER BY md."Marital_Status";
```

Data Output Messages Notifications

	Marital_Status character varying (12)	Most_Popular_ad_method text	Least_Popular_ad_method text
1	Divorced	Twitter	Brochure
2	Married	Instagram	Brochure
3	Single	Bulkmail	Brochure
4	Together	Twitter	Brochure
5	Widow	Twitter	Brochure

Which advertising method is most effective by country

```
SELECT
  md."Country",
  -- Most Effective Ad Method--
  CASE
    WHEN GREATEST(
      SUM(ad."Bulkmail_ad"::INT * md."Count_success"),
      SUM(ad."Twitter_ad"::INT * md."Count_success"),
      SUM(ad."Instagram_ad"::INT * md."Count_success"),
      SUM(ad."Facebook_ad"::INT * md."Count_success"),
      SUM(ad."Brochure_ad"::INT * md."Count_success")
    ) = SUM(ad."Bulkmail_ad"::INT * md."Count_success") THEN 'Bulkmail'
    WHEN GREATEST(
      SUM(ad."Bulkmail_ad"::INT * md."Count_success"),
      SUM(ad."Twitter_ad"::INT * md."Count_success"),
      SUM(ad."Instagram_ad"::INT * md."Count_success"),
      SUM(ad."Facebook_ad"::INT * md."Count_success"),
      SUM(ad."Brochure_ad"::INT * md."Count_success")
    ) = SUM(ad."Twitter_ad"::INT * md."Count_success") THEN 'Twitter'
    WHEN GREATEST(
```

```
      SUM(ad."Bulkmail_ad"::INT * md."Count_success"),
      SUM(ad."Twitter_ad"::INT * md."Count_success"),
      SUM(ad."Instagram_ad"::INT * md."Count_success"),
      SUM(ad."Facebook_ad"::INT * md."Count_success"),
      SUM(ad."Brochure_ad"::INT * md."Count_success")
    ) = SUM(ad."Instagram_ad"::INT * md."Count_success") THEN 'Instagram'
   WHEN GREATEST(
      SUM(ad."Bulkmail_ad"::INT * md."Count_success"),
      SUM(ad."Twitter_ad"::INT * md."Count_success"),
      SUM(ad."Instagram_ad"::INT * md."Count_success"),
      SUM(ad."Facebook_ad"::INT * md."Count_success"),
      SUM(ad."Brochure_ad"::INT * md."Count_success")
    ) = SUM(ad."Facebook_ad"::INT * md."Count_success") THEN 'Facebook'
    WHEN GREATEST(
      SUM(ad."Bulkmail_ad"::INT * md."Count_success"),
      SUM(ad."Twitter_ad"::INT * md."Count_success"),
      SUM(ad."Instagram_ad"::INT * md."Count_success"),
      SUM(ad."Facebook_ad"::INT * md."Count_success"),
      SUM(ad."Brochure_ad"::INT * md."Count_success")
    ) = SUM(ad."Brochure_ad"::INT * md."Count_success") THEN 'Brochure'
      ELSE 'None'
END AS "Most_Effective_Ad_Method",
-- Least Effective Ad Method
CASE
    WHEN LEAST(
      SUM(ad."Bulkmail_ad"::INT * md."Count_success"),
      SUM(ad."Twitter_ad"::INT * md."Count_success"),
      SUM(ad."Instagram_ad"::INT * md."Count_success"),
      SUM(ad."Facebook_ad"::INT * md."Count_success"),
      SUM(ad."Brochure_ad"::INT * md."Count_success")
    ) = SUM(ad."Bulkmail_ad"::INT * md."Count_success") THEN 'Bulkmail'
```

```
      WHEN LEAST(
        SUM(ad."Bulkmail_ad"::INT * md."Count_success"),
        SUM(ad."Twitter_ad"::INT * md."Count_success"),
        SUM(ad."Instagram_ad"::INT * md."Count_success"),
        SUM(ad."Facebook_ad"::INT * md."Count_success"),
        SUM(ad."Brochure_ad"::INT * md."Count_success")
      ) = SUM(ad."Twitter_ad"::INT * md."Count_success") THEN 'Twitter'
      WHEN LEAST(
        SUM(ad."Bulkmail_ad"::INT * md."Count_success"),
        SUM(ad."Twitter_ad"::INT * md."Count_success"),
        SUM(ad."Instagram_ad"::INT * md."Count_success"),
        SUM(ad."Facebook_ad"::INT * md."Count_success"),
        SUM(ad."Brochure_ad"::INT * md."Count_success")
      ) = SUM(ad."Instagram_ad"::INT * md."Count_success") THEN 'Instagram'
      WHEN LEAST(
        SUM(ad."Bulkmail_ad"::INT * md."Count_success"),
        SUM(ad."Twitter_ad"::INT * md."Count_success"),
        SUM(ad."Instagram_ad"::INT * md."Count_success"),
        SUM(ad."Facebook_ad"::INT * md."Count_success"),
        SUM(ad."Brochure_ad"::INT * md."Count_success")
      ) = SUM(ad."Facebook_ad"::INT * md."Count_success") THEN 'Facebook'
      WHEN LEAST(
        SUM(ad."Bulkmail_ad"::INT * md."Count_success"),
        SUM(ad."Twitter_ad"::INT * md."Count_success"),
        SUM(ad."Instagram_ad"::INT * md."Count_success"),
        SUM(ad."Facebook_ad"::INT * md."Count_success"),
        SUM(ad."Brochure_ad"::INT * md."Count_success")
      ) = SUM(ad."Brochure_ad"::INT * md."Count_success") THEN 'Brochure'
      ELSE 'None'
    END AS "Least_Effective_Ad_Method"
FROM "Marketing_Data" md
```

```
INNER JOIN "ad_data" ad ON md."ID" = ad."ID"
GROUP BY md."Country"
ORDER BY md."Country"
```

	Country character varying (6)	Most_Effective_Ad_Method text	Least_Effective_Ad_Method text
1	AUS	Instagram	Brochure
2	CA	Instagram	Brochure
3	GER	Twitter	Brochure
4	IND	Bulkmail	Brochure
5	ME	Bulkmail	Twitter
6	SA	Instagram	Brochure
7	SP	Instagram	Brochure
8	US	Facebook	Brochure

Which advertising method resulted in the highest increase in average spend

```
--add new column ;Media to ad_data--
ALTER TABLE public."ad_data"
ADD COLUMN "Media" VARCHAR(30);

SELECT * FROM public."ad_data";

--set Media as Bulkmail if "Bulkmail_ad"=1--
UPDATE public."ad_data"
SET "Media" = 'BulkMail'
WHERE "Bulkmail_ad"::int = 1;

--set Media as Twitter if "Twitter_ad"=1--
```

```
UPDATE public."ad_data"
SET "Media" = 'Twitter'
WHERE "Twitter_ad"::int = 1;

--set Media as Instagram if "Instagram_ad"=1--
UPDATE public."ad_data"
SET "Media" = 'Instagram'
WHERE "Instagram_ad"::int = 1;

--set Media as Facebook if "Facebook_ad"=1--
UPDATE public."ad_data"
SET "Media" = 'Facebook'
WHERE "Facebook_ad"::int = 1;

--set Media as Brochure if "Brochure_ad"=1--
UPDATE public."ad_data"
SET "Media" = 'Brochure'
WHERE "Brochure_ad"::int = 1;

--if "Media" = Null then set to'None'--
UPDATE public."ad_data"
SET "Media" = 'None'
WHERE "Brochure_ad"::int=0 AND "Twitter_ad"::int=0 AND "Instagram_ad"::int=0
AND "Facebook_ad"::int=0 AND "Brochure_ad"::int=0;

--select all media ads where customers were exposed to at least one--
--This doesn't take into account those customers who were exposed to multiple ad--
--it takes the last one checked as the media--
SELECT "Media" AS "Media_type",
  COUNT("ID") AS "Total_Customers",
  ROUND(AVG("Age"),0) AS "Avg_Age",
```

```
  ROUND(AVG("Income_($)"),2) AS "Avg_Income",
  ROUND(AVG("Total_Spend"),2) AS "Avg_Spend",
  ROUND(AVG("Alcohol"),2) AS "Avg_Alcohol",
  ROUND(AVG("Vegetables"),2) AS "Avg_veg",
  ROUND(AVG("Meat"),2) AS "Avg_Meat",
  ROUND(AVG("Fish"),2) AS "Avg Fish",
  ROUND(AVG("Chocolate"),2) AS "Avg Chocolate",
  ROUND(AVG("Commodities"),2) AS "Avg Commodities"
FROM public."Marketing_Data"
INNER JOIN public."ad_data" USING("ID")
GROUP BY "Media_type"
ORDER BY "Avg_Spend" DESC;
```

Showing rows: 1 to 5 Page No: 1 of 1

	Media_type character varying (100)	Total_Customers bigint	Avg_Age numeric	Avg_Income numeric	Avg_Spend numeric	Avg_Alcohol numeric	Avg_veg numeric	Avg_Meat numeric	Avg Fish numeric	Avg Chocolate numeric	Avg Commodities numeric
1	Instagram	77	56	80912.77	1609.26	852.01	49.69	464.17	67.51	56.96	66.87
2	Facebook	119	55	78582.20	1540.84	744.23	58.37	451.82	94.70	68.06	77.18
3	Brochure	26	55	71451.58	1370.19	897.31	24.46	260.04	38.96	31.96	70.88
4	Twitter	78	60	59363.26	848.14	564.85	19.41	132.35	25.82	17.63	35.12
5	None	1708	55	48243.35	531.75	228.56	23.40	134.83	32.60	23.39	40.06